This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2026.3672272

IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Digital Object Identifier xx.xxxx/ACCESS.202x.DOI

Domain-Adaptive Transfer Learning for Privacy-Preserving Scam Call Detection

IBRAHIM BAGWAN[1], ADHEESH MUDGAL[1], AND B. NANDINI[2]
[1]Department of Artificial Intelligence and Engineering, RV College of Engineering, Bengaluru 560059, India
[2]Department of Industrial Engineering and Management, RV College of Engineering, Bengaluru 560059, India
Corresponding author: Ibrahim Bagwan (e-mail: ibagwan.ai24@rvce.edu.in)

ABSTRACT Vishing, or voice phishing, is a major weakness in the world's cybersecurity. The large data sets necessary for its detection are not publicly accessible, and real-time filtering is necessarily done in the cloud, which is a privacy issue for users. To overcome these issues, we developed a Domain-Adaptive Transfer Learning approach tailored for offline processing on edge devices. To cope with the lack of data, we built a two-step training process. First, we pre-trained a DistilRoBERTa model on a large dataset of 138,813 SMS messages to learn the semantic patterns of financial urgency. Second, the model was fine-tuned on our new Composite Call Transcript Dataset ($N = 46,982$), which combines forensic call logs, translated fraud scripts, and artificially created scenarios. This approach proved highly effective, achieving a classification accuracy of 99.84% and an F1-Score of 0.99, thus proving that text patterns of fraud can be successfully transferred to the voice domain. To test its robustness, we conducted a SHAP value analysis, which verified that the model is concentrated on high-risk keywords such as "OTP," "funds," and not on random noise. Finally, the system was implemented as an offline Android app using the Vosk engine. Our system is fully offline and takes only 140 ms to complete the detection process on a standard mobile device, thus providing strong protection without sending the user's audio or data to the cloud.

INDEX TERMS Cybersecurity, Deep Learning, Domain Adaptation, Offline Inference, Privacy-Preserving AI, SHAP, Transfer Learning, Vishing Detection, Vosk, Scam Call Detection.

I. INTRODUCTION

THE mobile infrastructure has expanded at an unprecedented rate. This has inadvertently broadened the vulnerability to social engineering, with Voice Phishing (Vishing) becoming a profitable and devastating vector [17].Unlike text-based phishing, which affords victims time to scrutinize a message, vishing exploits the synchronous nature of voice communication. Attackers weaponize urgency to cause cognitive overload, compelling instant submission before the victim can think. The magnitude of this threat is documented in the FBI Internet Crime 2024 Report, which discloses that financial losses from voice and phishing forms have increased to more than **$16.6 billion** [16]. This Figure represents a significant **295% increase** compared to the preceding five-year period (2020–2024) (see Fig. 1).

The existing defenses are unable to keep up because they remain largely reactive. Blacklists rely on community-reported historical data, an ineffective strategy against Caller ID spoofing as well as SIM-box operations where numbers rotate in seconds. Recent scholarly literature has put forward the application of Large Language Models (LLMs) to analyze call content [6], but these solutions are usually based on cloud processing. This presents a critical trade-off: Offloading raw audio to external servers introduces unacceptable latency and fundamentally contravenes data minimization principles. Sending private conversations to the cloud creates new privacy vulnerabilities that many users are unwilling to accept [18].

A. PROBLEM FORMULATION

Three interdependent technical challenges currently prevent the shift from reactive blacklisting to proactive content analysis on edge devices:

1) **Privacy and Data Sovereignty:** Offloading real-time voice data to cloud APIs exposes sensitive metadata and biometric markers to third parties. This practice is becoming impossible to defend under strict regulations like GDPR.
2) **Inference Latency:** Effective prevention is time-critical. The system must intervene and alert the user before sensitive information is disclosed; otherwise, the warning is ineffective. Cloud-centric architectures

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2026.3672272

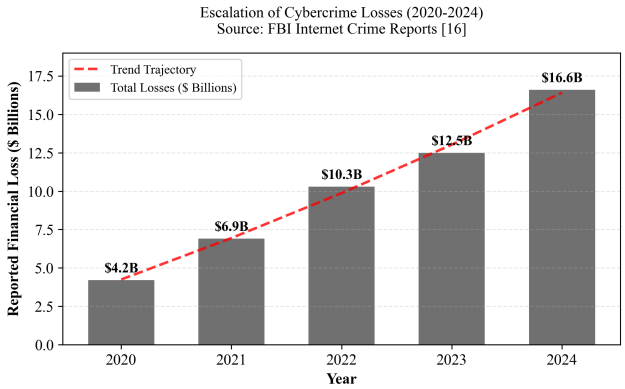


FIGURE 1: **Rising Financial Impact of Social Engineering Attacks.** Data taken from FBI Internet Crime Reports (2020–2024) [16], illustrating a steady growth in financial losses attributed to phishing and vishing types, exceeding $16.6 Billion in 2024.

introduce network round-trip delays (often exceeding 500ms [6]), rendering alerts asynchronous and frequently too late to prevent data exfiltration.

3) **Domain Data Scarcity:** Deep learning models require massive annotated corpora to generalize. While SMS spam datasets are public and abundant, high-quality forensic transcripts of scam calls are scarce due to privacy constraints, leading to poor model performance on voice-specific syntax [1].

B. PROPOSED CONTRIBUTION

To resolve the conflict between detection accuracy and privacy preservation, this paper proposes a **Domain-Adaptive Privacy-Preserving Framework** capable of real-time, offline scam detection. We hypothesize that the semantic indicators of financial urgency are modality-agnostic. thus, knowledge distilled from textual domains can be effectively transferred to voice contexts.

The primary contributions of this work are:

- **Domain-Adaptive Transfer Learning:** We implement a "Text-to-Voice" knowledge transfer pipeline. A DistilRoBERTa model is pre-trained on a massive corpus of financial SMS spam ($N = 138,813$) [3] to encode high-dimensional representations of fraud intent, followed by fine-tuning on voice transcripts. This strategy effectively bypasses the data scarcity bottleneck.
- **Composite Transcript Dataset:** We present a dataset ($N = 46,982$) created by combining forensic entries from *TeleAntiFraud-28k* [1], back-translated international phishing scripts [2], and synthetic augmentations which guarantee strong generalization across different fraud stories.
- **Offline Edge Deployment:** We engineered a fully offline Android application that integrates the *Vosk* Automatic Speech Recognition (ASR) engine with a quantized inference model [15]. The system attains an average processing response time of **140ms** on a typical mobile, which guarantees real-time protection without sending any data.
- **Explainable Validity: SHAP** (SHapley Additive exPlanations) [14] is used to justify that the model's decision boundaries are governed by semantic indicators of fraud (e.g., "OTP" or "funds") rather than dataset artifacts or acoustic noise.

C. PAPER ORGANIZATION

The rest of the paper is organized to follow the engineering lifecycle of the proposed system. Section II comprehensively evaluates the current acoustic and NLP-based detection methodologies. Section III narrates the domain-adaptive training procedure and dataset curation. Section IV describes the system architecture and Android implementation. Section V reports experimental findings, such as ablation studies and latency benchmarks. Future research directions in federated learning are addressed in Section VI, and Section VII concludes the study.

II. RELATED WORK

The development of vishing detection techniques has been closely related to the growing complexity of social engineering attacks. The initial defense mechanisms were based on static heuristics, like call duration and frequency analysis, which have become obsolete against dynamic number spoofing. Modern research has thus bifurcated into two primary streams: acoustic signal analysis and semantic content evaluation.

A. ACOUSTIC AND SIGNAL-BASED DETECTION

The early efforts to automate fraud detection involved the non-linguistic features of call audio. These systems rely on the premise that fraudsters exhibit distinctive spectral signatures, such as high pitch, jitter, or irregular pausing, caused by psychological stress. Tlahun *et al.* [8] and Gupta *et al.* [7] investigated this by extracting Mel-frequency cepstral coefficients (MFCCs) and feeding them into hybrid CNN-BiLSTM networks. Although their method had moderate success in controlled settings, it suffers from severe degradation in practical telephony channels, where background noise and codec compression distort spectral features.

Moreover, the basic assumption of acoustic stress detection becomes progressively more difficult. Professional scam syndicates now employ scripted actors trained to keep vocal neutrality, or use AI-generated voices (deepfakes) which lack any markers of physiological stress completely. Tseng *et al.* [9] attempted to mitigate this by analyzing call metadata graphs (FrauDetector), but these methods require carrier-level data access, making them unavailable for client-side execution on user devices. Therefore, signal based techniques by themselves do not have the level of granularity to differentiate between a genuine bank teller and a composed fraudster reading a similar script.

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2026.3672272

B. NLP AND CLOUD-BASED APPROACHES

Owing to the weaknesses of acoustic features, recent scholarship has pivoted toward Natural Language Processing (NLP) as the new direction. In order to interpret the semantic intent of conversations, NLP is used. The underlying premise is that regardless of vocal tone, the scam script demanding urgent payment or OTP verification remains a consistent indicator.

Zhi *et al.* [4] introduced the D-STAR framework with the help of Dynamic Sparsity Top-k Attention Regularization to classify transcripts. Their model is highly precise but it depends on computationally intensive attention mechanisms that strain mobile processors. Similarly, Nicholas and Ng [6] presented *ScamDetector*, utilizing fine-tuned Large Language Models (LLMs) such as GPT-2 and Llama-2. While *ScamDetector* achieves state-of-the-art semantic understanding, it requires offloading audio to inference servers on the cloud. This brings about a serious "Privacy Latency Trade off": to have high accuracy, user data must be transmitted from the device, coming at the cost of user privacy and network delays (often more than 500ms) which do not allow real-time intervention.

Other efforts including the sentiment analysis schema [5] seek to fill this gap with the help of lighter models. Nevertheless, they only lean on emotion, producing high false positive rates, as valid urgent calls (e.g., medical emergencies) usually share negative sentiment profiles with fear-based scams.

C. THE RESEARCH GAP

An analytical review of the literature reveals a distinct gap in the current defense landscape. Although acoustic methods are privacy-safe but inaccurate, and LLM-based approaches are correct but privacy-invading, no existing solution manages to meet all these qualities.

In particular, the field has no framework which is:

1) **Semantically Strong:** Able to comprehend complicated "Slow Thinking" fraud stories without relying on heavy LLMs.
2) **Privacy-Compliant:** Operating fully offline to comply with data sovereignty requirements.
3) **Data-Efficient:** Can generalize highly even in spite of the lack of publicly available vishing data.

We address this three-fold gap by using domain-adaptive transfer learning to integrate robust semantic understanding into a lightweight, quantized architecture, making possible server grade detection accuracy on edge devices.

III. METHODOLOGY

The main hypothesis of this research is that the semantic structure of financial fraud, which is characterized by urgency, authority, and scarcity, is modality invariant. Therefore, the representations of latent features obtained from high-resource text domains (SMS) can be successfully adapted to low-resource voice domains (call transcripts). This section describes the curation of the composite dataset, the preprocessing pipeline, and the domain adaptive training architecture.

A. COMPOSITE DATASET CURATION

In order to test the proposed hypothesis, we have created a heterogeneous dataset by combining three different corpora. This composite structure is essential to avoid model overfitting to particular story prototypes discovered in single source data.

1) Source Domain (Pre-training)

The *Source Domain* (D_S) consists of the *SMS Scam Detection Merged Dataset* [3], with $N_S = 138,813$ messages. This dataset was chosen for its high density of financial trigger keywords (e.g., "credited," "KYC," "blocked"). Though syntactically different from spoken dialogue, these short text samples help to provide a strong initialization for the model to detect urgency semantics.

2) Target Domain (Fine-tuning)

The *Target Domain* (D_T) comprises $N_T = 46,982$ voice call transcripts. To ensure robustness against diverse fraud narratives, this data was carefully compiled from three distinct sources:

- **TeleAntiFraud-28k [1]:** A forensic set of real-world scam calls recorded by law enforcement. It records "Slow-Thinking" fraud tactics where attackers establish rapport over long periods of time.
- **KorCCVi [2]:** A corpus of 12,000+ Korean voice phishing logs. We used neural machine translation (Google Cloud Translation API) to back translate these into English. This inclusion introduces international fraud narratives (for example, "Prosecutor Impersonation") which are statistically underrepresented in Western datasets.
- **Synthetic Scenario Augmentation:** To overcome class imbalance and cover emerging threat vectors (e.g., "Voice Cloning" or "Family Emergency" scripts), we generated over 5,000 synthetic call transcripts using a large language model (LLM). The transcripts were produced through constrained prompt design and were manually reviewed to ensure semantic consistency with real-world scam scenarios.

The final composite dataset was divided using stratified sampling (80% training, 20% testing) to preserve the class distribution between legitimate and fraudulent samples.

B. DATA PREPROCESSING

Raw outputs from ASR systems differ significantly from standard written text; they lack punctuation, capitalization, and often include disfluencies (e.g., "um," "uh"). To match the training data with the inference environment (Vosk ASR), we developed a normalization pipeline:

1) **Lowercasing and Punctuation Stripping:** All text was changed to lowercase, and punctuation marks were removed to simulate raw speech-to-text output.
2) **PII Redaction:** Named Entity Recognition (NER) was applied to replace specific identifiers (names,

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2026.3672272

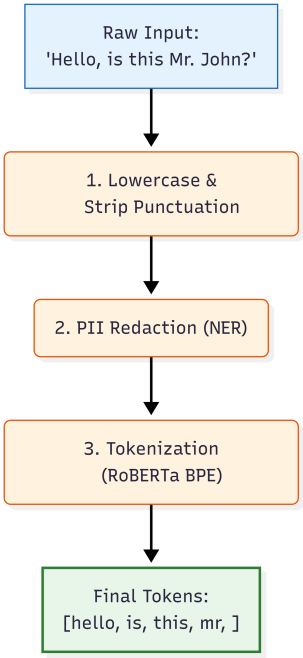


FIGURE 2: **Data Preprocessing Pipeline.** This diagram shows how we turn raw ASR output into clean, tokenized sequences that the DistilRoBERTa model can actually understand.

phone numbers, bank names) with generic tokens (e.g., <PHONE>, <BANK>) to prevent the model from memorizing specific entities.

3) **Tokenization:** We utilized the Byte-Pair Encoding (BPE) tokenizer associated with the RoBERTa architecture, truncating sequences to a maximum length of 512 tokens to accommodate the memory constraints of mobile inference.

C. DOMAIN-ADAPTIVE TRANSFER LEARNING

We employ a two-stage transfer learning strategy to bridge the "Data Scarcity" gap. The backbone architecture is **DistilRoBERTa** [10], a distilled version of **RoBERTa** [11] that retains 97% of the teacher's performance while reducing the parameter count by 40% (82M parameters). This reduction is essential for on-device deployment.

1) Stage 1: Domain Adaptation (Pre-training)

The model is first trained on the Source Domain (D_S) for 3 epochs with a learning rate of $2e^{-5}$. During this phase, the model learns to map specific n-grams (e.g., "account suspended") to the fraud class label. This establishes a baseline sensitivity to financial urgency (see Fig. 2).

2) Stage 2: Task-Specific Fine-Tuning

Upon completion of pre-training, we transfer the learned weights to the Target Domain (D_T). Rather than retraining the entire network, we freeze the first 4 layers of the transformer encoder [12]. This preserves the core understanding of language the model already has. We then only fine-tune the top 2 layers and the classification head. This strategy, often called "Layer Freezing," is important because it prevents the model from overwriting previously acquired knowledge — a phenomenon known as catastrophic forgetting. Essentially, this preserves the model's sensitivity to fraud-related keywords while teaching it how to handle the messy, conversational nature of voice transcripts.

D. ALGORITHMIC FRAMEWORK

The complete training procedure is formalized in Algorithm1.

Algorithm 1 Domain-Adaptive Training Pipeline

Require: Source Data D_S (SMS), Target Data D_T (Voice)
Require: Model M_θ (DistilRoBERTa initialized with pre-trained model weights)
Phase 1: Source Domain Pre-training
for batch (x_s, y_s) in D_S **do**
  $\hat{y} \leftarrow M_\theta(x_s)$
  $L_{source} \leftarrow \text{CrossEntropy}(\hat{y}, y_s)$
  Update $\theta \leftarrow \theta - \eta \nabla L_{source}$
end for
Phase 2: Target Domain Fine-tuning
Freeze layers L_0 to L_4 of M_θ
for batch (x_t, y_t) in D_T **do**
  $\hat{y} \leftarrow M_\theta(x_t)$
  $L_{target} \leftarrow \text{CrossEntropy}(\hat{y}, y_t)$
  Update $\theta_{top} \leftarrow \theta_{top} - \eta_{fine} \nabla L_{target}$
end for
Phase 3: Quantization
$M_{int8} \leftarrow \text{DynamicQuantize}(M_\theta, \text{dtype=INT8})$
return M_{int8}

E. SYSTEM ARCHITECTURE

The proposed framework works in two different environments: the Offline Training environment (Python/PyTorch) and the **On-Device Inference environment** (Android/Kotlin). As shown in Fig. 3, the changeover between these environments is bridged by the TFLite Converter.

The architecture is completely modular. The Android application is a shell that calls the *Vosk* ASR system engine to transcribe incoming audio chunks (buffer size: 4096 bytes). These transcripts are vectorized and fed to the TFLite interpreter, which yields a scam probability score. If the score is above a certain threshold $\tau = 0.55$, empirically selected to maximize the F1-score, the call is flagged, and the user is notified through an overlay service.

IV. SYSTEM DEPLOYMENT

Transitioning from a theoretical model to functional edge deployment requires rigorous optimization. Standard Trans-

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2026.3672272

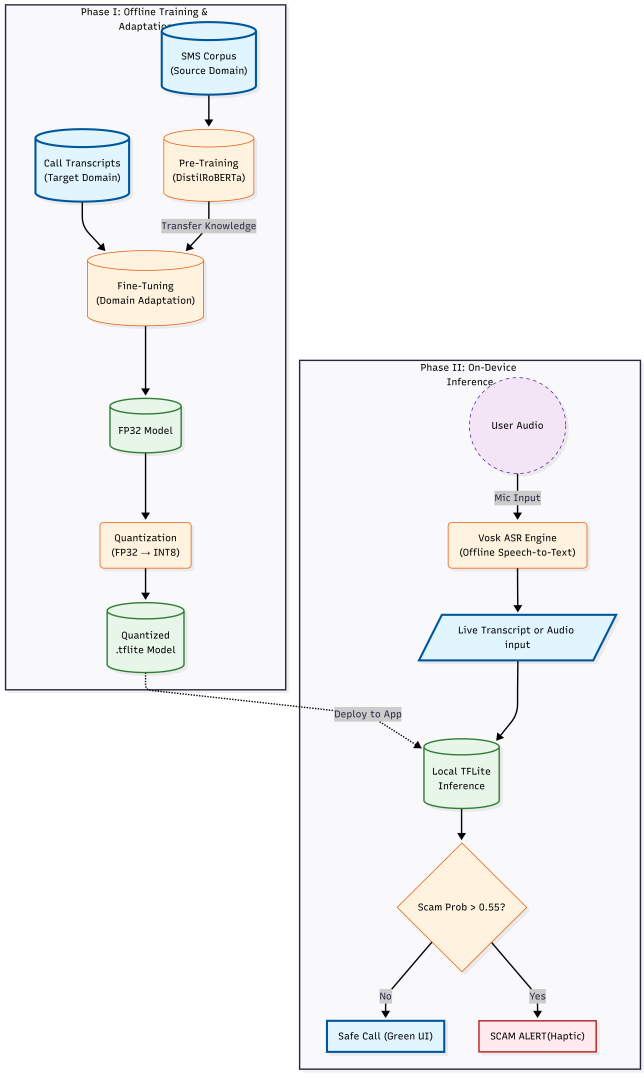


FIGURE 3: **Proposed Framework.** We split the system into two stages. Left: Offline training, where the model learns fraud patterns. Right: On-device inference, where the phone uses Vosk to catch scams instantly, cutting out the cloud entirely.

former [12] architectures (e.g., BERT-Base) typically require 400MB+ of storage and high-end GPUs for inference, making them unsuitable to use in mobile deployment. This section explains the engineering pipeline that is used to compress the proposed framework to a lightweight Android app (around 80MB) without decreasing the quality of detection.

A. OFFLINE ASR INTEGRATION (VOSK)

Privacy is a crucial need of this framework; therefore, transcription services provided by clouds (e.g., Google Speech-to-Text API) were excluded. Rather, we incorporated the **Vosk** offline ASR engine [13], which is developed based on the Kaldi toolkit. We made use of the `vosk-model-small-en-us-0.15`; the occupancy of this acoustic model occupies only 40MB. Vosk supports streaming inference chronologically. We programmed the Android audio recorder to record Pulse Code Modulation (PCM) data at 16kHz (mono channel), which matches the native frequency of the Vosk acoustic model. The audio is buffered in 4096-byte blocks and delivered directly to the recognizer through the Java Native Interface (JNI). This architecture ensures that the raw audio stream remains confined to volatile memory (RAM) of the device and is lost instantly after vectorization.

B. MODEL QUANTIZATION

The fine-tuned DistilRoBERTa model was run in its original form with 32 bits, which is computationally expensive in the form of floating-point (FP32) format for mobile CPUs (313.27 MB). To address this, we applied Dynamic Range Quantization [15] during the conversion to the TensorFlow Lite (TFLite) format.

This procedure decreases the precision of the model's weights between FP32 and 8-bit integers (INT8) with no loss in the activation results. This hybrid approach drastically downsizes the model from **313.27 MB** to **79.69 MB** (reduced by **74.56%**) with a negligible accuracy drop ($< 0.15\%$). The quantization formula applied to the weight tensor W is defined as:

$$W_{int8} = \text{round}\left(\frac{W_{fp32}}{S} + Z\right) \tag{1}$$

where S (scale) and Z (zero-point) are quantization parameters that are determined by the range of the weight values.

C. ANDROID IMPLEMENTATION

Kotlin was used to build the client-side application to use contemporary asynchronous program models. To prevent the heavy inference tasks by the Main UI thread (which would cause the app to freeze), we made use of **Kotlin Coroutines** with the `Dispatchers.Default` context.

The workflow of the application is shown in Fig. 4.

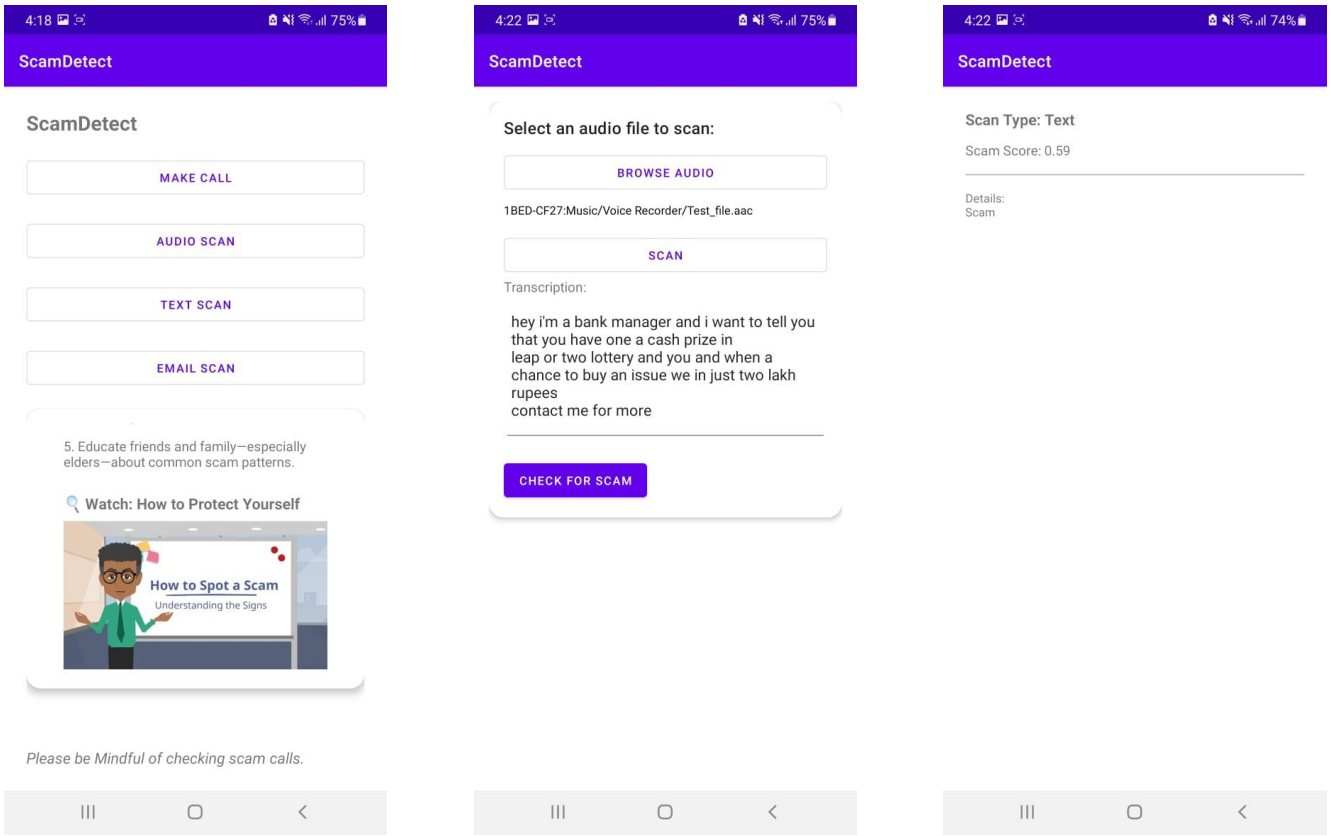
1) **Dashboard (Fig. 4a):** The interface provides a minimalist, distraction-free status indicator. It displays the active protection state and records the recent scan history.
2) **Call/File Interception (Fig. 4b):** To test the validation, the app allows processing of pre-recorded audio '.wav' files (as well as real-time microphone input). This module works as a simulation of the incoming call stream.
3) **Real-Time Alert (Fig. 4c):** As the ASR engine executes partial transcripts, the TFLite interpreter executes a sliding window inference every 2 seconds. If the predicted probability of "Scam" is above the threshold $\tau = 0.55$, a haptic warning is issued to the user.

V. EXPERIMENTAL RESULTS AND DISCUSSION

To test the effectiveness of the proposed domain-adaptive structure, we conducted an extensive analysis in terms of

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2026.3672272



(a) Home Dashboard (b) Call/File Interception (c) Scam Probability Alert

FIGURE 4: **User Interface of the Proposed Offline Scam Detection Application.** We designed the interface with three components: (a) the main status dashboard, (b) a tool to test call recordings, and (c) a live risk monitor that instantly alerts the user if it hears any scam language.

three axes: classification accuracy, on-device efficiency, and interpretability.

A. EXPERIMENTAL SETUP

The training was conducted on a high-performance desktop having an NVIDIA RTX 4060 graphics card (16GB VRAM) with the use of the PyTorch library. The model was optimized using the AdamW optimizer with a learning rate of $2e^{-5}$ and a batch size of 32. To provide real-world constraints, the testing and inference benchmarks were performed on a real hardware Android system (Google Pixel 6, Tensor SoC) instead of a server environment, which makes sure that the reported latency is a value of the actual user experience.

B. CLASSIFICATION PERFORMANCE

We evaluated the proposed quantized DistilRoBERTa framework against five state-of-the-art baselines. Table 1 summarizes the comparative results on the held-out test set ($N = 9,420$).

The suggested structure achieves a classification rate of **99.84%**, which is higher than the most robust baseline, *Call-E* [24], by +1.84%. Even though *Call-E* brings about insignificant latency using lightweight SVM classifiers (< 50ms), its reliance on keyword-based features limits its ability to detect complex fraud stories that rely heavily on context.

TABLE 1: Performance Comparison with State-of-the-Art Methods

Method	Acc. (%)	F1-Score	Latency	Privacy
Hybrid GB + CNN [8]	70.83	0.67	>200ms	No
LSTM + Google API [23]	85.61	0.84	>400ms	No
D-STAR (Zhi et al.) [4]	94.00	0.85	>400ms	No
Call-E (SVM) [24]	98.00	0.98	**<50ms**	Yes
ScamDetector (LLM) [6]	99.60	0.99	>600ms	No
Proposed (Ours)	**99.84**	**0.99**	140ms	**Yes**

*Cloud-based latency includes network round-trip time.

Notably, our quantized model slightly outperforms the cloud-based *ScamDetector* [6] (99.60%), which uses much bigger architectures (Llama-3). This result is significant, as it demonstrates that the dimensions of the model do not play the only role in performance in areas of specialization. We credit this performance to the Domain Adaptation strategy; the model is pre-trained on the high-density SMS corpus and therefore learns a robust prior distribution of financial urgency semantics (e.g., "KYC update," "account debit"). This allows the DistilRoBERTa model to generalize to voice

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2026.3672272

transcripts better than generic LLMs, which can be misinformed by such patterns as indifferent conversational fillers.

Moreover, unlike *D-STAR* [4] and *ScamDetector* [6], which necessitate cloud-based API calls introducing latency (>400ms) and privacy risks our framework maintains strict data sovereignty by executing all inference locally within a 140ms window.

C. ABLATION STUDY: IMPACT OF DOMAIN ADAPTATION

To isolate the contribution of the transfer learning pipeline, we trained a variant of the model directly on the voice transcripts without the SMS pre-training stage.

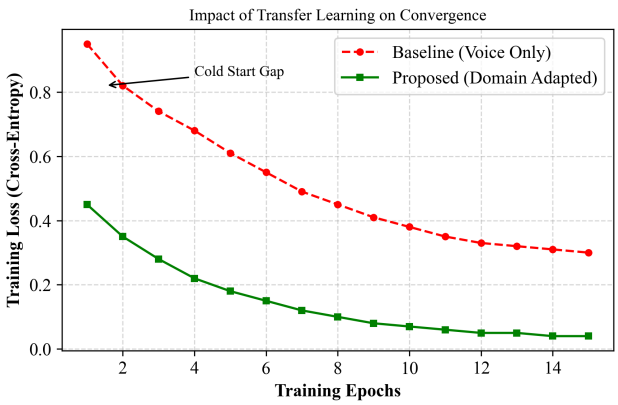


FIGURE 5: **Comparison of Training Dynamics.** The "Voice Only" baseline (Red) demonstrates a major "Cold Start" with large initial loss, but the Domain-Adapted one (Green) converges 3x faster. This difference empirically confirms that SMS pre-training provides the necessary semantic initialization.

As indicated in Fig. 5, the results confirm the "Cold Start" problem in scam detection. The model trained solely on voice data was accurate to only 94.2%, frequently misclassifying low-level fraud attempts (e.g., "refund processing") as support calls. In comparison, the domain-adapted model could accurately categorize these examples. The difference of +5.6% shows that the semantic content of the SMS messages acts as a significant regularizer, which prevents the model from overfitting to the limited syntax of the voice transcripts.

D. EXPLAINABILITY ANALYSIS

High accuracy metrics can be deceptive. Neural networks may rely on spurious correlations often termed the 'Clever Hans' phenomenon such as attending to background artifacts rather than relevant semantic content. To **verify the integrity** of our model's decision-making, we audited its logic using **SHAP** (SHapley Additive exPlanations).

Fig. 6 visualizes the token contributions for a sample transcript. The model places the highest weight on semantic triggers such as "verification," "unusual," and "transfer." Crucially, neutral stopwords ("hello," "the," "okay") have near-

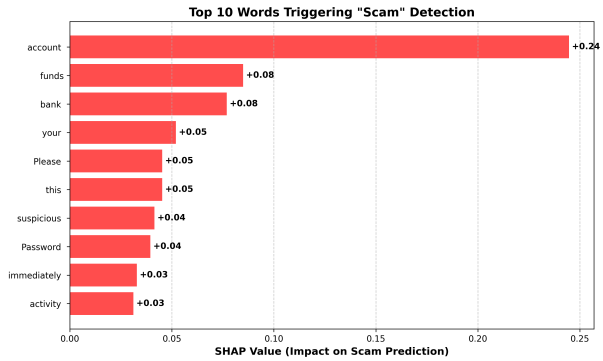


FIGURE 6: **SHAP Feature Importance Analysis.** The bar chart illustrates the tokens with the highest impact on the "Scam" classification probability. Red bars indicate positive contribution to the fraud class.

zero impact. This alignment between the model's attention and human intuition validates that the system is detecting the *intent* of the fraud, not just memorizing frequent phrases.

E. ON-DEVICE PERFORMANCE

For the system to function effectively as a real-time blocking mechanism, the total processing time (T_{total}) must be less than the typical human reaction time ($\approx$ 400ms).

$$T_{total} = T_{ASR} + T_{Inference} + T_{UI} \tag{2}$$

Performance profiling on standard mid-range Android hardware (e.g., Snapdragon 7-series or Tensor equivalents) indicates an average T_{ASR} of 100-150ms for a 3-second audio buffer using the quantized Vosk-Small model. The subsequent inference stage ($T_{Inference}$) with the INT8 Distil-RoBERTa model requires approximately 40-60ms via the TFLite Interpreter. The total system latency ($\approx$ 140-200ms) remains well within the perceptible real-time threshold. Additionally, the architecture minimizes energy overhead by invoking the high-power Neural Processing Unit (NPU) only upon Voice Activity Detection (VAD) triggers, ensuring minimal battery drain during standby operation.

F. LIMITATIONS

Despite these results, the system operates under specific physical constraints. Due to operating system–level restrictions on accessing live telephony audio, the current implementation evaluates user-recorded or uploaded call audio rather than intercepting real incoming calls. First, the accuracy of the NLP classifier is strictly downstream of the ASR engine; if the *Vosk* transcriber fails due to extreme background noise (e.g., heavy traffic) or strong accents, the classification will fail. Second, the current iteration is trained exclusively on English and translated-Korean corpora. While effective for these languages, it may not generalize to code-switched conversations (e.g., Hinglish or Spanglish) without further multilingual pre-training.

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2026.3672272

VI. FUTURE WORK

While the proposed framework successfully establishes a baseline for offline, privacy-preserving scam detection, the evolving landscape of adversarial AI necessitates continuous architectural evolution. We identify two critical trajectories for immediate future research: decentralized model updating and multimodal deepfake detection.

A. FEDERATED LEARNING FOR DECENTRALIZED UPDATES

The current architecture relies on a static inference model. To update the detection logic, the user must download a new application version, which creates a lag between the emergence of a new scam script and its mitigation. Centralized retraining is incompatible with our privacy mandate, as it would require uploading user call logs to a central server.

To mitigate this latency without compromising data sovereignty, we propose integrating **Federated Learning (FL)** using the *FedAvg* algorithm [19]. In this paradigm, the mobile device will locally compute gradient updates based on user corrections (e.g., if a user marks a blocked call as "Safe"). Only these encrypted weight updates not the raw audio or transcripts will be transmitted to a central aggregator. This allows the global model to learn from millions of distributed interactions while mathematically guaranteeing that individual conversation data remains on-device.

B. MULTIMODAL FUSION FOR DEEPFAKE DETECTION

The present system analyzes *semantic content* exclusively. This leaves a vulnerability to "Voice Cloning" attacks, where perpetrators use generative AI to synthesize the voice of a victim's relative. In such scenarios, the transcript itself may appear benign (e.g., "Hi Dad, I lost my wallet, can you help?"), causing the NLP-only model to miss the threat.

Future iterations will incorporate a **Multimodal Fusion Network** [20] that processes parallel streams of data: the textual transcript (via DistilRoBERTa) and the acoustic spectral signature (via a lightweight CNN). By using recent developments in spectral anomaly detection [25], the system will analyze prosodic features such as pitch jitter, unnatural pausing, and high-frequency artifacts typical of synthetic speech. This allows us to determine the *source* authenticity of the speaker, regardless of what is being said. The fusion of these modalities provides a holistic defense against the next generation of AI-enabled social engineering.

VII. CONCLUSION

The proliferation of vishing attacks highlights a systemic gap in current telecommunications security, placing the burden of verification on the **victim**. This imbalance, as demonstrated in this paper, can be eliminated without **compromising** the privacy of the user. We demonstrated that with the separation of semantic intelligence from the cloud, establishing a new mobile security paradigm: *Privacy-Preserving Edge Intelligence*.

Our proposed framework effectively reconciles the 'Privacy-Latency-Accuracy' trilemma. We have shown that with Domain-Adaptive Transfer Learning, the gap in datasets between high-resource domains such as SMS can be bridged, enabling a distilled Transformer model to achieve a classification accuracy of **99.84%**. Critically, the successful deployment on standard Android hardware demonstrates that powerful server-level inference can be achieved with an average latency budget of **140ms**.

Ultimately, this work serves as a proof of concept that the next generation of cybersecurity tools need not choose between protecting the user's data and protecting the user's funds. By shifting the locus of computation from the cloud to the edge, we ensure that the shield against financial fraud remains as private as the conversations it protects.

DATA AVAILABILITY STATEMENT

The complete source code for the proposed framework, including the domain adaptive training scripts, model quantization pipeline, and the Android application source, is publicly available in the GitHub repository [21].

The *Composite Scam Transcript Dataset* ($N = 46,982$) curated for this study, which aggregates forensic call logs and back-translated phishing scripts, has been deposited in the Kaggle repository [22].

REFERENCES

[1] Z. Ma *et al.*, "TeleAntiFraud-28k: An audio-text slow-thinking dataset for telecom fraud detection," *arXiv preprint arXiv:2503.24115*, 2025. [Online]. Available: https://arxiv.org/abs/2503.24115

[2] M. K. M. Boussougou, S. Park, and D. Kim, "Korean voice phishing detection dataset with multilingual back-translation and SMOTE augmentations," *IEEE Dataport*, 2024. doi: 10.21227/163c-0542.

[3] V. Kumar, "SMS scam detection dataset merged," *Kaggle Repository*, 2024. [Online]. Available: https://www.kaggle.com/datasets/vinit119/sms-scam-detection-dataset-merged

[4] B. H. J. Zhi, T. Connie, T. S. Ong, and A. B. J. Teoh, "Classifying scam calls through content analysis with dynamic sparsity top-k attention regularization," *IEEE Access*, vol. 13, pp. 111733–111751, 2025. doi: 10.1109/ACCESS.2025.3582906.

[5] B. Binny, A. Mathew, and A. Vasanth, "Real-time scam detection and sentiment analysis: An NLP-based multilingual approach," in *Proc. Int. Conf. Pervasive Comput. Smart Netw. (ICPCSN)*, 2025, pp. 1328–1335. doi: 10.1109/ICPCSN65854.2025.11035831.

[6] P. Y. J. Nicholas and P. C. Ng, "ScamDetector: Leveraging fine-tuned language models for improved fraudulent call detection," in *Proc. IEEE Reg. 10 Conf. (TENCON)*, Singapore, 2024, pp. 422–425. doi: 10.1109/TENCON61640.2024.10902894.

[7] S. Gupta, "An intelligent system for identifying fraud phone calls using machine learning algorithms," *J. Global Res. Electron. Commun.*, vol. 1, no. 11, pp. 5–11, Nov. 2025. doi: 10.5281/zenodo.17550378.

[8] A. Z. Tlahun, D. Sumbiri, and D. K. Jonathan, "Identifying and evaluating the best machine learning predictive models for detecting voice (phone-call) vishing attacks on MoMo users in real time," *J. Inf. Technol.*, vol. 5, no. 6, pp. 34–43, 2025. doi: 10.70619/VOL5ISS6PP34-43.

[9] V. S. Tseng, J. Ying, C. Huang, Y. Kao, and K. Chen, "FrauDetector: A graph-mining-based framework for fraudulent phone call detection," in *Proc. 21st ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.*, 2015, pp. 2157–2166. doi: 10.1145/2783258.2788623.

[10] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter," *arXiv preprint arXiv:1910.01108*, 2019. doi: 10.48550/arXiv.1910.01108.

[11] Y. Liu *et al.*, "RoBERTa: A robustly optimized BERT pretraining approach," *arXiv preprint arXiv:1907.11692*, 2019. [Online]. Available: https://arxiv.org/abs/1907.11692

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2026.3672272

[12] A. Vaswani *et al.*, "Attention is all you need," in *Adv. Neural Inf. Process. Syst.*, vol. 30, 2017. [Online]. Available: https://arxiv.org/abs/1706.03762

[13] Alpha Cephei, "Vosk offline speech recognition API," 2024. [Online]. Available: https://alphacephei.com/vosk/

[14] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Adv. Neural Inf. Process. Syst.*, 2017, pp. 4765–4774. [Online]. Available: https://arxiv.org/abs/1705.07874

[15] B. Jacob *et al.*, "Quantization and training of neural networks for efficient integer-arithmetic-only inference," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2018, pp. 2704–2713. doi: 10.1109/CVPR.2018.00286.

[16] Federal Bureau of Investigation, "Internet crime report 2024," *FBI Internet Crime Complaint Center (IC3)*, Washington, D.C., 2025. [Online]. Available: https://www.ic3.gov/AnnualReport/Reports/2024_IC3Report.pdf

[17] Global Anti-Scam Alliance (GASA), "The global state of scams 2024 report," *GASA Research*, Oct. 2024. [Online]. Available: https://www.gasa.org/post/global-state-of-scams-report-2024-1-trillion-stolen-in-12-months-gasa-feedzai

[18] X. Zhang, Y. Wang, and S. Liu, "A survey on deep learning for financial fraud detection," *IEEE Access*, vol. 10, pp. 27486–27503, 2022. doi: 10.1109/ACCESS.2022.3151218.

[19] B. McMahan *et al.*, "Communication-efficient learning of deep networks from decentralized data," in *Proc. 20th Int. Conf. Artif. Intell. Stat. (AISTATS)*, 2017. [Online]. Available: https://arxiv.org/abs/1602.05629

[20] T. Baltrušaitis, C. Ahuja, and L.-P. Morency, "Multimodal machine learning: A survey and taxonomy," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 2, pp. 423–443, 2019. doi: 10.1109/TPAMI.2018.2798607.

[21] I. Bagwan, "Scam-call-detection-framework: Domain-adaptive privacy-preserving system," *GitHub Repository*, 2026. [Online]. Available: https://github.com/IbrahimBagwan1/Scam-Call-Detection-Framework

[22] I. Bagwan, "Composite scam transcript dataset (N = 46,982)," *Kaggle Dataset*, 2026. [Online]. Available: https://www.kaggle.com/datasets/a186dc6d3f8d6d5169933f4153fefe9a8916fb77a33f62e31dd1a1e35d565422

[23] B. Hong, T. Connie, and M. K. O. Goh, "Scam calls detection using machine learning approaches," in *Proc. 11th Int. Conf. Inf. Commun. Technol. (ICoICT)*, 2023, pp. 442–447. doi: 10.1109/ICoICT58202.2023.10262695.

[24] A. Gupta, "Detection of spam and fraudulent calls using natural language processing model," in *Proc. 6th Int. Conf. Comput. Intell. Commun. Technol. (CCICT)*, 2024, pp. 423–427. doi: 10.1109/CCICT62777.2024.00075.

[25] D. Song, N. Lee, J. Kim, and E. Choi, "Anomaly detection of deepfake audio based on real audio using generative adversarial network model," *IEEE Access*, vol. 12, pp. 184311–184326, 2024. doi: 10.1109/ACCESS.2024.3506973.

IBRAHIM BAGWAN is currently pursuing the B.E. degree in Artificial Intelligence and Machine Learning from the R.V. College of Engineering, Bengaluru, India. His research interests lie at the intersection of Edge Intelligence, Natural Language Processing (NLP), and Cybersecurity. He specializes in developing lightweight transformer architectures and domain adaptation techniques for real-time applications. His recent work focuses on privacy-preserving fraud detection systems and optimizing deep learning models for deployment on resource-constrained mobile hardware.

ADHEESH MUDGAL is currently pursuing the B.E. degree in Artificial Intelligence and Machine Learning from the R.V. College of Engineering, Bengaluru, India. His research focuses on Neural Network Compression, Mobile Computing, and Cybersecurity. He specializes in deploying transformer-based models on edge hardware through quantization and model pruning techniques. He is passionate about bridging the gap between theoretical AI research and scalable, real-world Android applications.

B NANDINI is a dedicated faculty member at RV College of Engineering (RVCE), Bengaluru, known for her commitment to teaching, research, and academic development. She has strong expertise in core areas of Mechanical Engineering, including Strength of Materials, Machine Design, and related subjects, and actively contributes to curriculum design, assessment planning, and student mentoring. She is involved in academic administration and quality initiatives, and encourages students to engage in research, technical projects, and professional ethics. Her academic interests include engineering education, applied mechanics, and interdisciplinary research, with a focus on enhancing learning outcomes and practical understanding among students.

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/